An Illumination Estimation and Compensation Method for Radiometric Correction of UAV Multispectral Images

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***Abstract*— The multispectral imaging of unmanned aerial vehicle (UAV) is often affected by the variation of illumination, resulting in serious spectral radiation distortion. Precise illumination estimation and compensation is a key step to carry out the radiometric correction on UAV multispectral images (MSIs), especially for the case without irradiance sensors. To accurately estimate the illumination for the radiometric correction, a physics-based illumination estimation and compensation method is proposed in this article. In the proposed method, an illumination estimation model is built based on the intraimage hypothesis on illumination consistency and the interimage hypothesis on reflectance consistency. This model is used to obtain the illumination irradiance of each one from numerous MSIs simultaneously. Then the influence of varying illumination can be alleviated with the estimated irradiance based on the physical imaging principle. To validate the effectiveness of the proposed method, numerical experiments are conducted on three UAV datasets acquired under cloudy weather. The experimental results demonstrate that the proposed method outperforms current methods and the normalized root mean square error (NRMSE) on the three datasets are noticeably reduced.**

***Index Terms*— Illumination compensation, illumination variation, multispectral image (MSI), unmanned aerial vehicle (UAV).**

I. Introduction

SPECTRAL imaging is an important means of Earth observation tasks since most materials exhibit unique spectral signatures when photons of light interact with surface materials [1]. The early imaging spectrometers are mostly based on the satellite platform or large aircraft, which are limited by low resolution, long revisiting times, and prohibitive costs [2]. Recently, there have been some development and improvement in the sensors and platforms, such as the miniaturization of optical payload systems [3] and the autonomous navigation of unmanned aerial vehicles (UAVs) [4]. Then the combination of miniaturized multispectral sensors and UAV platforms is proposed as a complement to the traditional way. This combination has gradually become the primary choice for near-Earth observation tasks due to the advantages of high accuracy, flexible implementation, and low cost. The application fields where the UAV multispectral images (MSIs) are generally employed include ecological monitoring [5], forestry investigation [6], agriculture management [7], etc.

The radiometric accuracy of MSI is influenced by several factors during the imaging process. It is an important issue in quantitative remote sensing to study how to alleviate the influence of external factors [8], [9]. For satellite and airborne platforms, atmospheric interference is the main source of radiometric errors for MSI [10]. As for UAV platforms, the illumination variation is the main source of radiometric errors for MSIs [11] because the influence of atmospheric attenuation is negligible due to the low altitude [12]. Considering that the altitude of the UAV is far below the cloud, the sharp illumination variations caused by the fluctuating cloud cover density inevitably are involved in radiometric errors. Besides, the continuous slow illumination variation caused by changes in the solar altitude angle is also a cause. According to the downwelling irradiance measurements of pyranometers described in [13], the irradiance can fluctuate between 200 and 350 W/m^2 during a single flight campaign under various illumination conditions. The influence of the varying illumination is illustrated in Fig. 1, where the brightness difference between images can be seen on the mosaic image in a linear stretch. So far, the most popular method of radiometric correction for UAV MSIs is the empirical line method (ELM), which derives the reflectance factor from known reflectance targets [7], [14]. The ELM, however, is not robust to illumination variations since the influence of illumination can only be eliminated when reflectance panels are visible in each image. This is not practical for large experiment areas [15]. Therefore, the compensation for the illumination variation is a distinctive challenge for UAV MSIs.

Recently, some attention has been paid to the illumination compensation for MSIs on UAV platforms, which can be categorized into the following two strategies: the *irradiance-sensor-oriented* and *data-driven* strategies. The main idea of the *irradiance-sensor-oriented* strategy is to exploit irradiance sensors to measure illumination irradiance directly and continuously and compensate for the impact of illumination variations with the physical imaging principle. Some researchers

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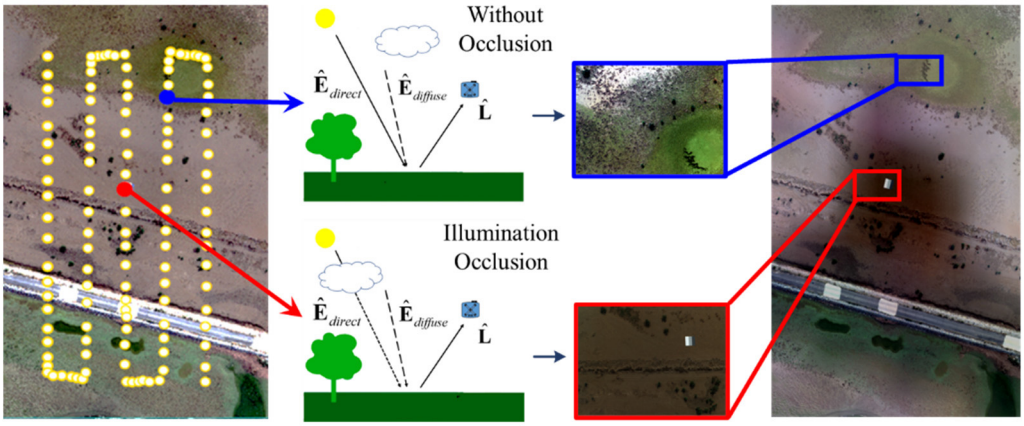


Fig. 1. Illustration of the influence of illumination variations on the mosaic MSI. $\hat{\mathbf{E}}_{\text{direct}}$ and $\hat{\mathbf{E}}_{\text{diffuse}}$ denote the direct irradiance and the scattered irradiance, respectively. $\hat{\mathbf{L}}$ denotes the at-sensor radiance of the multispectral camera.

used commercial off-the-shelf spectrometers to measure the real-time irradiance on the ground or onboard during the flight [13], [16]. For example, Burkart et al. [16] exploited the Ocean STS-VIS Spectrometer as a ground unit to measure the in situ irradiance. Suomalainen et al. [17] mounted an Ocean Optics USB2000+ spectrometer with cosine response on UAV for real-time solar irradiance. Since commercial spectrometers are too heavy for UAV platforms and the bands need to be realigned to cameras, the self-designed and miniaturized irradiance sensors are proposed to meet the requirements of the weight and band [18], [19]. Suomalainen et al. [18] combined photodiodes and well-selected spectral filters to measure direct and diffuse irradiance. Late, MicaSense also designed an onboard down-welling light sensor (DLS) for RedEdge and Altum cameras to acquire the irradiance in real time [19]. The measurements of irradiance sensors are subject to systematic errors due to the variation of UAV attitude and the altitude difference between the UAV and the ground. In addition, the irradiance-sensor-oriented methods are inapplicable to cameras without irradiance sensors.

Instead of using irradiance sensors, the *data-driven* algorithms are based on the linear correction model between images and use statistical information to homogenize the images under different illumination conditions [20], [21], [22]. For example, Honkavaara et al. [20] first proposed a relative correction method called the radiometric block adjustment method. They formulated a linear correction model between images and estimated linear coefficients according to the tie points between overlapping images. Subsequently, Shin et al. [21] developed an algorithm called relative calibration by the optimal path (RCOP). They selected an optical path for each image to the reference image for illumination compensation in order to minimize the accumulated error. Lately, Kizel et al. [22] extended ELM to the case of multiple images, referred to as the generalized ELM (GELM). Although GELM can obtain the linear coefficient of multiple images simultaneously, it has only been validated on four MSIs. In summary, data-driven algorithms are typically based on linear correction models between images and are not suitable for numerous MSIs due to the large accumulated error and weak dependence on physical imaging principles. As there are usually hundreds of MSIs in a single flight, it is necessary to propose an illumination compensation method for the case of numerous MSIs.

In this article, a physics-based illumination estimation and compensation method are proposed in the absence of irradiance sensors for numerous MSIs. Our motivation is intuitive. We can combine the advantages of the two strategies by substituting the irradiance sensor measurement with the estimated irradiance of each image. Thus, we propose here an irradiance-sensor-independent and physics-based method for illumination compensation. Our method consists of three parts: 1) preliminary processing; 2) illumination estimation model; and 3) optimization and solution. The preliminary processing is to provide proper tie points and reference points. Then we formulate the illumination estimation model for numerous MSIs under the interimage and intraimage hypotheses. In the last step, the irradiance of numerous MSIs can be estimated simultaneously, and the influence of varying illumination is alleviated with estimated irradiances.

The main contributions of this article are as follows:

1) First, an illumination estimation model is built to implement the illumination estimation for numerous MSIs. In this model, we introduce the intraimage hypothesis on illumination consistency and the interimage hypothesis on reflectance consistency so that the irradiance of numerous MSIs can be estimated simultaneously. An important advantage is that it can minimize the accumulated errors on marginal images since the hundreds of irradiances are obtained in a single mathematical operation instead of stepwise iterations. To our knowledge, it is the first attempt at illumination estimation for UAV MSIs.
2) Second, a physics-based illumination estimation and compensation method is proposed without irradiance sensors. This method stems from the physical imaging principle and alleviates the influence of the varying illumination with the estimated illumination irradiance.

The remaining part of the article proceeds as follows. In Section II, we formulate the proposed illumination and

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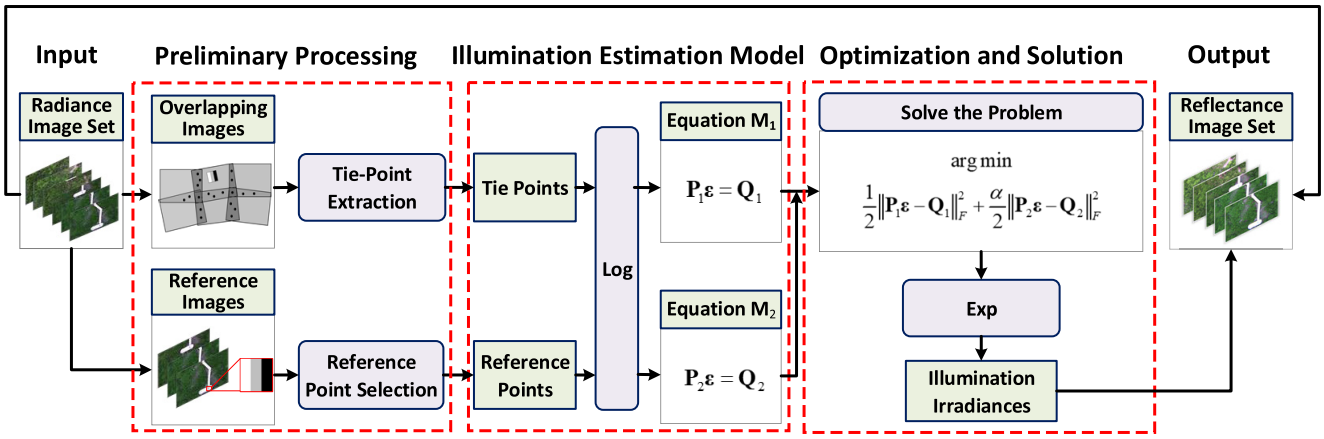


Fig. 2. Workflow for the proposed illumination estimation and compensation method.

compensation method in detail. The experiment results conducted on three field-experiment datasets are presented in Section III followed by the conclusion remarks in Section IV.

II. Illumination Compensation Method

To compensate for the influence of illumination variations without irradiance sensors, we propose a physics-based illumination estimation and compensation method for numerous MSIs to alleviate the influence of illumination variation during campaigns. The workflow for the proposed method is shown in Fig. 2.

A. Preliminary Processing

According to the imaging model in (3), the radiance is a function of the illumination and reflectance factors. Because of the inequality between the number of known and unknown parameters, there is no unique solution for separating illumination and reflectance factors from radiance images alone. Thus, the illumination estimation for numerous MSIs is an ill-posed problem. In this article, we introduce priors on tie points and reference points to reduce the number of unknown parameters and make the problem solvable. The before use of tie point enables building connections between overlapping images, while the reference points introduce before reflectance on panels. The tie-point extraction and reference point selection are conducted as preliminary processing.

1) Tie-Point Extraction: This step is executed on each pair of overlapping images. Here we use a common way that adopts the scale-invariant feature transform (SIFT) algorithm [23] to extract feature matches and the random sample consensus (RANSAC) algorithm [24] to eliminate the wrong matches. Then we get the initial tie points.

There might be a difference in the radiance of tie points due to the varying imaging angles. It is necessary to filter the outliers for high accuracy. For each pair of overlapping images, we first calculate the normalized root mean square error (NRMSE) between tie points and eliminate the largest 30% of them. Let Δ_{mean} and σ denote the mean value and standard deviation of the radiance difference of all tie points, respectively. The qualified tie points should meet the following condition:

$$|\Delta_s(\lambda) - \Delta_{\text{mean}}(\lambda)| \leq \eta \cdot \sigma(\lambda) \tag{1}$$

where $\Delta_s(\lambda)$ denotes the radiance difference in the band λ of the sth pair of tie points, and η is the threshold.

The setting and analysis of the parameter η are described in Section III-D.

2) Reference Point Selection: The reference point selection step is executed on the reference image, which contains reflectance panels in the field of view. There are usually 3–9 reference images in a flight campaign. In each reference image, the reference point should be manually selected for each known panel. We label an area and substitute its average for a single pixel for less error.

Except for the two steps above, we take the log of images and panel spectra to transform them into the logarithmic domain. The reason is described in Section II-B.

B. Illumination Estimation Model

Let $\mathbf{I} \in \mathbb{R}^{W \times H \times P}$ denote an MSI, and W, H, P denote its width, height, and band number. The dataset with N MSIs is denoted as $\mathbf{\Theta} = \{\mathbf{I}_1, \ldots, \mathbf{I}_N\}$.

Note: We use the symbol of hat $\wedge$ to denote nonlogarithmic variables and drop the hat to denote logarithmic variables so that we could distinguish them from each other.

According to the atmospheric radiative transfer models (RTM) [25], [26], [27], the physical imaging model is expressed as

$$\hat{\mathbf{L}} = \tau \hat{\mathbf{L}}_{\text{reflected}} + \hat{\mathbf{L}}_{\text{path}} \tag{2}$$

where τ denotes the atmospheric upward transmittance factor, and $\hat{\mathbf{L}}, \hat{\mathbf{L}}_{\text{reflected}}, \hat{\mathbf{L}}_{\text{path}}$ denote the at-sensor radiance, target-reflected radiance, and atmospheric path radiance.

The influence of atmospheric attenuation is negligible in this work since the flight altitude is usually less than 120 m [28], [29]. Therefore, τ is set to 1 and $\hat{\mathbf{L}}_{\text{path}}$ is negligible. The simplified physical imaging model for UAV is given as

$$\hat{\mathbf{L}} = \hat{\mathbf{E}} \circ \hat{\mathbf{R}} / \pi \tag{3}$$

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where $\hat{\mathbf{E}}$ denotes the illumination irradiance, $\hat{\mathbf{R}}$ denotes the reflectance factor, and the operator $\circ$ denotes the element-wise multiplication.

Due to the element-wise multiplication operator, (3) cannot be rewritten in matrix form. Therefore, we transform (3) into the logarithmic domain. Here we denote the logarithms of $\hat{\mathbf{L}}, \hat{\mathbf{E}}, \hat{\mathbf{R}}$ by $\mathbf{L}, \mathbf{E}, \mathbf{R}$, respectively. The logarithmic and matrix form of (3) can be expressed as

$$\mathbf{L} = \mathbf{E} + \mathbf{R} - \boldsymbol{\pi} \tag{4}$$

where $\boldsymbol{\pi} \in \mathbb{R}^{P\times 1}$ is given as (5).

$$\underset{P\times 1}{\boldsymbol{\pi}} = \left[\log(\pi), \ldots, \log(\pi)\right]^T. \tag{5}$$

Since illumination estimation for hundreds of MSIs simultaneously is an ill-posed problem, we introduce the intraimage hypothesis on illumination consistency and the interimage hypothesis on reflectance consistency to reduce the number of unknown parameters and make it solvable.

1) Intraimage Hypothesis: We assume that all pixels in a single image share the same illumination.

$$\mathbf{E}_k(x_i, y_i) = \mathbf{E}_k\left(x_j, y_j\right) \quad \forall i, j \in \mathbf{I}_k \tag{6}$$

where (x_i, y_i), (x_j, y_j) denote the coordinate of the pixel i, j in image $\mathbf{I}_k$, and $\mathbf{E}_k$ denotes the irradiance of $\mathbf{I}_k$ in log.

The constancy of illumination within an image depends on two parts: 1) the proportion of the spatial extent of the image and that of cloud shadow and 2) the velocity of shadow edges caused by moving clouds. The spatial extent of cloud shadow is larger than that of a single UAV MSI. Taking RedEdge-MX Dual [34] as an example, the coverage area of an image is about 60 × 50 m at 80 m altitude. The average length of cloud shadow in the moving direction is 803 m based on the irradiance data of a total of 12 000 shading periods in 2011–2014 [35]. Although a cloud shadow edge might appear in the camera's field of view, it will disappear in a few seconds due to the high shadow speed (13.2 m/s on average [35]) and slow camera capture period (about 2 s per image). Thus, the environment illumination of most images is consistent throughout the image. *Note*: This study does not consider cast shadows caused by tall objects.

2) Interimage Hypothesis: We assume the reflectance factors of each pair of tie points in overlapping images are equal.

$$\mathbf{R}_l\left(x_s^l, y_s^l\right) = \mathbf{R}_r\left(x_s^r, y_s^r\right), \quad r \neq l \tag{7}$$

where the subscript r, l denote the index of overlapping images, (x_s^l, y_s^l), (x_s^r, y_s^r) denote the coordinate of the sth pair of tie points in $\mathbf{I}_r$ and $\mathbf{I}_l$.

As described in [17] and [36], the bidirectional reflectance distribution function (BRDF) effect is not obvious in cloudy conditions, and the introduction of BRDF factors complicates the solution processing greatly. Therefore, we do not take the BRDF into consideration in this method.

According to (4), the interimage hypothesis in (7) can be rewritten as

$$\mathbf{E}_r - \mathbf{E}_l = \mathbf{L}_r\left(x_s^r, y_s^r\right) - \mathbf{L}_l\left(x_s^l, y_s^l\right) \tag{8}$$

where $\mathbf{L}_r(x_s^r, y_s^r), \mathbf{L}_l(x_s^l, y_s^l)$ denote the at-sensor radiance of the corresponding pixels of the sth pair of tie points between $\mathbf{I}_r$ and $\mathbf{I}_l$ in the log domain.

Based on the hypotheses mentioned above, we utilize the tie points extracted in Section II-A to build a connection between overlapping images. Let m denote the number of tie points for the entire image set $\boldsymbol{\Theta}$. Then we conduct the equation system $\mathbf{M}_1$ with $(m \times P)$ linear equations

$$\mathbf{M}_1 = \left\{ \begin{array}{l} \mathbf{E}_{r_1} - \mathbf{E}_{l_1} = \mathbf{L}_{r_1}\left(x_1^{r_1}, y_1^{r_1}\right) - \mathbf{L}_{l_1}\left(x_1^{l_1}, y_1^{l_1}\right) \\ \mathbf{E}_{r_2} - \mathbf{E}_{l_2} = \mathbf{L}_{r_2}\left(x_2^{r_2}, y_2^{r_2}\right) - \mathbf{L}_{l_2}\left(x_2^{l_2}, y_2^{l_2}\right) \\ \vdots \\ \mathbf{E}_{r_m} - \mathbf{E}_{l_m} = \mathbf{L}_{r_m}\left(x_m^{r_m}, y_m^{r_m}\right) - \mathbf{L}_{l_m}\left(x_m^{l_m}, y_m^{l_m}\right) \end{array} \right\} \tag{9}$$

where the subscript r_t, l_t denote the index of overlapping images that the tth pair of tie points for $\boldsymbol{\Theta}$ are extracted from.

According to the intraimage hypothesis, we utilize the reference points selected in Section II-A to introduce extra priors on the reflectance. Let n denote the total number of reference points in the image set $\boldsymbol{\Theta}$. We conduct the equation system $\mathbf{M}_2$ with $(n \times P)$ linear equations

$$\mathbf{M}_2 = \left\{ \begin{array}{l} \mathbf{E}_{k_1} = \mathbf{L}_{k_1}\left(x_1^{k_1}, y_1^{k_1}\right) - \mathbf{R}^1 + \boldsymbol{\pi} \\ \mathbf{E}_{k_2} = \mathbf{L}_{k_2}\left(x_2^{k_2}, y_2^{k_2}\right) - \mathbf{R}^2 + \boldsymbol{\pi} \\ \vdots \\ \mathbf{E}_{k_n} = \mathbf{L}_{k_n}\left(x_n^{k_n}, y_n^{k_n}\right) - \mathbf{R}^n + \boldsymbol{\pi} \end{array} \right\} \tag{10}$$

where the subscript k_t denotes the index of the image that the tth reference point belongs to, $\mathbf{R}^t$ denotes the reflectance factor of the tth reference point in the log domain.

Let $\boldsymbol{\varepsilon} \in \mathbb{R}^{(N\times P)\times 1}$ denotes the illumination irradiance of all images, which is given as

$$\underset{(N\times P)\times 1}{\boldsymbol{\varepsilon}} = [\mathbf{E}_1, \ldots, \mathbf{E}_N]^T. \tag{11}$$

The matrix form of $\mathbf{M}_1, \mathbf{M}_2$ can be expressed as

$$\mathbf{A}_1\boldsymbol{\varepsilon} = \mathbf{B}_1 \tag{12}$$

$$\mathbf{A}_2\boldsymbol{\varepsilon} = \mathbf{B}_2 \tag{13}$$

where $\mathbf{A}_1 \in \mathbb{R}^{(m\times P)\times(N\times P)}$, $\mathbf{A}_2 \in \mathbb{R}^{(n\times P)\times(N\times P)}$ denote the coefficient matrix of $\boldsymbol{\varepsilon}$ in $\mathbf{M}_1$ and $\mathbf{M}_2$, and $\mathbf{B}_1 \in \mathbb{R}^{(m\times P)\times 1}, \mathbf{B}_2 \in \mathbb{R}^{(n\times P)\times 1}$ denote the right-hand vector of $\mathbf{M}_1$ and $\mathbf{M}_2$.

Since they are large matrices, we cannot give the explicit form of the matrix $\mathbf{A}_1$ and $\mathbf{A}_2$. For better understanding, we take a pair of tie points and a reference point as examples and show their corresponding rows.

For a pair of tie points between the first and third image, the corresponding rows in $\mathbf{A}_1$ are given as

$$\left[\mathbf{I}_{P\times P} \quad \mathbf{0}_{P\times P} \quad -\mathbf{I}_{P\times P} \quad \mathbf{0}_{P\times P} \quad \cdots \quad \mathbf{0}_{P\times P}\right]_{P\times(N\times P)}. \tag{14}$$

As for a reference point in the 2nd image, the corresponding rows in $\mathbf{A}_2$ are given as

$$\left[\mathbf{0}_{P\times P} \quad \mathbf{I}_{P\times P} \quad \mathbf{0}_{P\times P} \quad \mathbf{0}_{P\times P} \quad \cdots \quad \mathbf{0}_{P\times P}\right]_{P\times(N\times P)}. \tag{15}$$

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To combine (12) and (13) in the solution process, we zero-pad the matrix $\mathbf{A}_1$ and $\mathbf{A}_2$ to keep their dimensionalities consistent.

$$\mathbf{P}_1 = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{0}_{(n\times P)\times(N\times P)} \end{bmatrix}_{[(m+n)\times P]\times(N\times P)} \tag{16}$$

$$\mathbf{P}_2 = \begin{bmatrix} \mathbf{0}_{(m\times P)\times(N\times P)} \\ \mathbf{A}_2 \end{bmatrix}_{[(m+n)\times P]\times(N\times P).} \tag{17}$$

Similarly, we zero-pad the matrix $\mathbf{B}_1$ and $\mathbf{B}_2$.

$$\mathbf{Q}_1 = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{0}_{(n\times P)\times 1} \end{bmatrix}_{[(m+n)\times P]\times 1}$$

$$\mathbf{Q}_2 = \begin{bmatrix} \mathbf{0}_{(m\times P)\times 1} \\ \mathbf{B}_2 \end{bmatrix}_{[(m+n)\times P]\times 1.} \tag{18}$$

Equations (12) and (13) are rewritten as

$$\mathbf{P_1}\boldsymbol{\varepsilon} = \mathbf{Q}_1 \tag{19}$$

$$\mathbf{P}_2\boldsymbol{\varepsilon} = \mathbf{Q}_2. \tag{20}$$

The simultaneous equations composed of (19) and (20) are overdetermined and have no unique solution since there are at least dozens of tie points between each pair of overlapping images. By using the Lagrange multipliers, a unique solution with minimum error is obtained. The formulation of the illumination estimation model is given as

$$U = \frac{1}{2}\|\mathbf{P}_1\boldsymbol{\varepsilon} - \mathbf{Q}_1\|_F^2 + \frac{\alpha}{2}\|\ \mathbf{P}_2\boldsymbol{\varepsilon} - \mathbf{Q}_2\|_F^2 \tag{21}$$

$$\boldsymbol{\varepsilon} = \arg\min_{\boldsymbol{\varepsilon}}\ U \tag{22}$$

where $\alpha \geq 0$ is a Lagrange multiplier, and the symbol $\|\mathbf{V}\|_F$ denotes the Frobenius norm of the vector $\mathbf{V}$.

C. Optimization and Solution

This section is to optimize and solve the illumination estimation model in (21) based on the intraimage and interimage hypotheses. In this way, we obtain the estimated irradiances and the retrieved reflectance images.

As the objective function in (21) is quadratic, the solution lies in the null points of its first partial derivatives.

Let

$$\partial U(\boldsymbol{\varepsilon}, \alpha)\big/\partial\boldsymbol{\varepsilon} = 0. \tag{23}$$

We can obtain the following linear equation system

$$\mathbf{P}_1^T(\mathbf{P}_1\boldsymbol{\varepsilon} - \mathbf{Q}_1) + \alpha\mathbf{P}_2^T(\mathbf{P}_2\boldsymbol{\varepsilon} - \mathbf{Q}_2) = \mathbf{0}. \tag{24}$$

The (24) could be simplified as

$$\big(\mathbf{P}_1^T\mathbf{P}_1 + \alpha\mathbf{P}_2^T\mathbf{P}_2\big)\boldsymbol{\varepsilon} = \mathbf{P}_1^T\mathbf{Q}_1 + \alpha\mathbf{P}_2^T\mathbf{Q}_2. \tag{25}$$

The illumination irradiance in the log domain of all images $\{\mathbf{E}_1, \mathbf{E}_2, \ldots, \mathbf{E}_N\}$ can be achieved simultaneously by solving the linear equation system in (25).

Transform $\mathbf{E}_k$ back into the nonlogarithmic domain

$$\hat{\mathbf{E}}_k = \exp(\mathbf{E}_k). \tag{26}$$

The kth reflectance image in $\boldsymbol{\Theta}$ could be retrieved based on the physical imaging model

$$\hat{\mathbf{R}}_k(x, y, \lambda) = \pi\hat{\mathbf{L}}_k(x, y, \lambda)\big/\hat{\mathbf{E}}_k(x, y, \lambda). \tag{27}$$

In summary, the whole procedure of the proposed illumination estimation and compensation is shown in Algorithm 1.

Algorithm 1 The Proposed Method

Input: MSI dataset in radiance $\{\hat{\mathbf{L}}_1, \ldots, \hat{\mathbf{L}}_N\}$, and the spectra of panels $\{\hat{\mathbf{R}}^1, \hat{\mathbf{R}}^2, \ldots, \hat{\mathbf{R}}^t\}$, parameter $\alpha = 0.07$.

Output: Multispectral image set in reflectance $(\hat{\mathbf{R}}_1, \ldots, \hat{\mathbf{R}}_N)$.

1: **Logarithm:** $\mathbf{L}_k = \log(\hat{\mathbf{L}}_k)$

2: **Tie-Points Extraction:**

Extract tie points for each pair of overlapping images with SIFT and RANSAC methods.

We get m pairs of tie point for the entire image set $\boldsymbol{\Theta}$.

3: **Reference Points Selection:**

Select reference points manually for reference images.

We get n reference points for the entire image set $\boldsymbol{\Theta}$.

4: **Illumination Estimation Model:**

Calculate the matrix $\mathbf{P}_1, \mathbf{Q}_1$ with m pairs of tie points according to Eq. (9) and (19).

Calculate the matrix $\mathbf{P}_2, \mathbf{Q}_2$ with n reference points according to Eq. (10) and (20).

5: **Solve the Equation:**

$$\big(\mathbf{P}_1^T\mathbf{P}_1 + \alpha\mathbf{P}_2^T\mathbf{P}_2\big)\boldsymbol{\varepsilon} = \mathbf{P}_1^T\mathbf{Q}_1 + \alpha\mathbf{P}_2^T\mathbf{Q}_2$$

Then we get the irradiance of all images $(\mathbf{E}_1, \ldots, \mathbf{E}_N)$.

6: **Exponentiation:**

$$\hat{\mathbf{E}}_k = \exp(\mathbf{E}_k)$$

7: **Reflectance Factor Retrieve:**

$$\hat{\mathbf{R}}_k(x, y, \lambda) = \pi\hat{\mathbf{L}}_k(x, y, \lambda)\big/\hat{\mathbf{E}}_k(x, y)$$

III. Experiments Results and Analysis

A. Data Description

To evaluate the effectiveness of our method, we carried out three field experiments and acquired corresponding datasets of MSIs. All flight campaigns are carried out in cloudy weather and last for at least 20 min to ensure that there are illumination variations during the flight. The details of the field experiments are described as follows:

1) HIT Campus: The first field experiment was carried out at the Harbin Institute of Technology (HIT), Harbin, China. We placed the calibrated tarp with three different reflective panels (56%, 30%, and 11% reflectance) as reference targets in the experimental area before the flight. Besides, the ground reflectance factors of several uniform targets, such as black fabric, roads, and carton boxes, were measured using PSR 3500+ Spectrometer. The position is marked in Fig. 3(a). The dataset was captured using the Micasense RedEdge-MX Dual camera carried by the DJI M300 RTK at 80 m altitude with nine strips. It contains 215 MSIs in ten bands with a GSD of 5.3 cm. The real-time irradiance of the blue band recorded by DLS is shown in Fig. 3(d).

2) ShanKou Mangrove: The second field experiment was carried out in ShanKou Mangrove (SKM) Nature Reserve in China. We also placed the calibrated tarp as reference targets in the experimental area before the flight. Besides, the ground reflectance factors of several uniform targets, such

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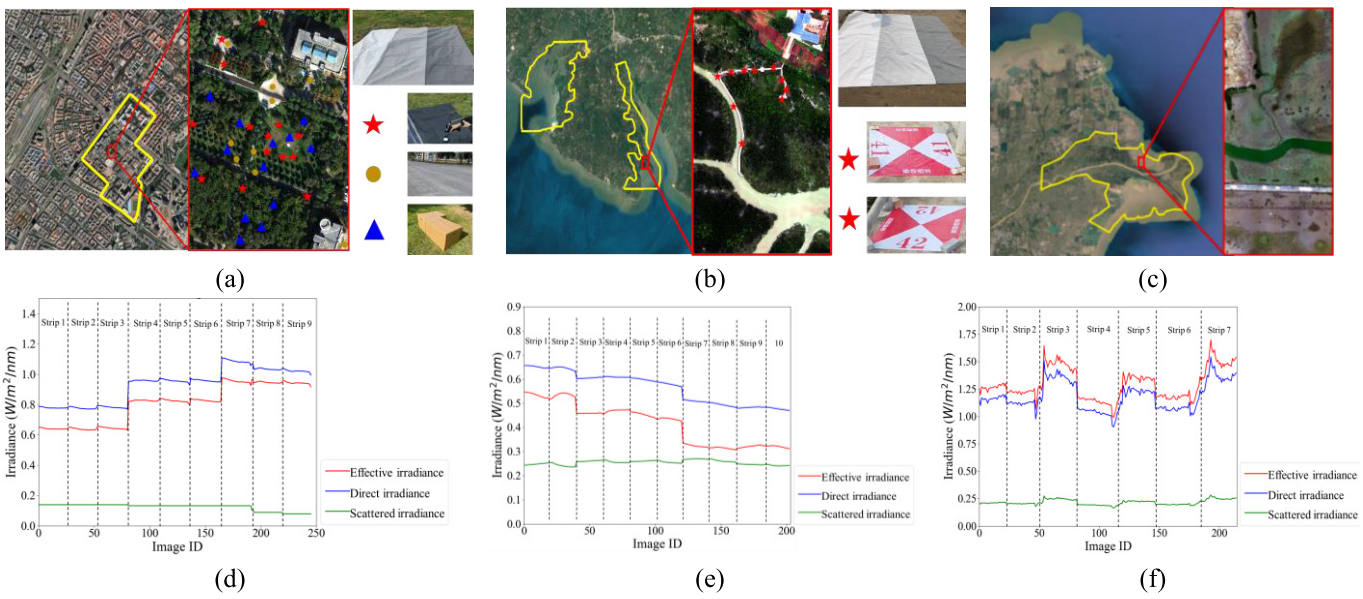


Fig. 3. (a)–(c) Field experiment location and ground target positions of the HIT, SKM, and YRD datasets, respectively. The campus or reserve is outlined in yellow, while the experiment area is marked with a red rectangle. (d)–(f) Real-time irradiances of the blue band recorded by DLS of the three datasets, respectively.

as aerial survey targets, were measured using PSR 3500+ Spectrometer. The positions are marked in Fig. 3(b). The dataset was captured using the Micasense RedEdge-MX Dual camera carried by the DJI M300 RTK at 50 m altitude with ten strips. It contains 203 MSIs in ten bands with a GSD of 4 cm. The real-time irradiance of the blue band recorded by DLS is shown in Fig. 3(e).

3) Yellow River Delta: This experiment was carried out in the Yellow River Delta (YRD) National Nature Reserve in China. We also placed the calibrated tarp as reference targets in the experimental area before the flight. Because of the lack of spectrometers, we did not measure ground reflectance factors. The dataset was captured using the Micasense RedEdge-M camera carried by the DJI M600 Pro at 50 m altitude with seven strips. It contains 215 MSIs in five bands with a GSD of 3.5 cm. The real-time irradiance of the blue band recorded by DLS is shown in Fig. 3(f).

B. Experiment Setting

1) Accuracy of Estimated Irradiances: As the proposed method can produce an estimated spectral irradiance for each image, we present the DLS irradiance time series and estimated irradiance time series of the three datasets. Then, using the DLS real-time irradiance as the truth, the average relative error of each dataset is given.

2) Effectiveness of Illumination Compensation: The ground-truth targets are usually few in number and cannot be evenly distributed across the entire experimental scene, the traditional evaluation method described in [21] and [22] with ground-truth targets is not suitable for assessing the effectiveness of illumination compensation. Thus, we propose a new way to verify the effectiveness of illumination compensation for the whole experiment scene. Here we use the MSIs corrected by Micasense's official method (DLS + calibrated tarps) [31] as reference data for no illumination variations. We then generate a mosaic image as the reference mosaic image. Since the influence of the varying illumination can be eliminated by the DLS irradiances, the reference mosaic image can be identified as illumination-independent. Then the difference between the corrected and reference mosaic image can indicate how much the illumination has influenced. Besides, the experiment results can be visualized with this reference mosaic image.

We use NRMSE to evaluate the difference in reflectance factors between the pixel in the corrected image and the corresponding pixel in the reference mosaic image

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{P}\sum_{i=1}^{P}\left[R(\lambda_i) - R_{\mathrm{valid}}(\lambda_i)\right]^2}}{|R_{\mathrm{valid}}|} \tag{28}$$

where R_{valid} and R denote the reflectance factor of the pixel in the reference mosaic image and its corresponding pixel in the corrected image, and $|\cdot|$ denotes the norm of a vector.

We visualize the results of each pixel with its NRMSE value based on the color slice. To analyze the effectiveness of illumination compensation of the reference and proposed methods, we calculate the mean and standard deviations (STD) of NRMSE for the entire image.

In this experiment, we choose ELM [7] to represent errors before the illumination compensation as it is not robust to the illumination variation, and we choose two data-driven methods as compared to methods, which are GELM [22] and RCOP [21].

3) Accuracy of Reflectance Factors: Using the spectrometer-based reflectance factors of uniform targets as the ground truth, we analyze the error of the corrected UAV MSIs. Here we choose the RMSE and NRMSE to represent the absolute and relative errors.

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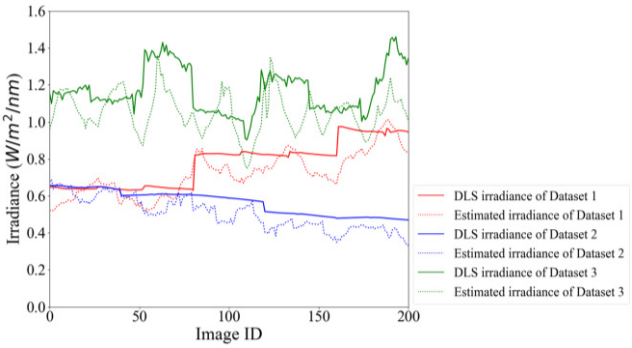


Fig. 4. DLS irradiance time series and estimated irradiance time series of three datasets.

TABLE I
MEAN AND STANDARD DEVIATION OF NRMSE FOR THE ENTIRE IMAGE ON THREE DATASETS

		ELM	GELM	RCOP	Proposed Method
HIT	MEAN (%)	22.22	37.91	17.01	**9.47**
	STD (%)	18.98	20.09	13.57	**8.85**
SKM	MEAN (%)	20.90	47.79	20.18	**7.28**
	STD (%)	14.24	32.69	11.85	**5.85**
YRD	MEAN (%)	16.38	14.66	14.81	**9.77**
	STD (%)	14.32	12.88	7.48	**6.80**

The setting and analysis of the parameter α in the proposed method are described in Section III-D. The parameters of reference methods consist of the corresponding references.

To eliminate the influence of the sensor factors, all experiments are performed on multispectral radiance images after radiometric calibration, which is based on the image processing tutorials from Micasense with the correction coefficients from the EXIF metadata [32], [33].

All methods were run with Python 3.6 on Windows 7. The CPU is Intel Core i7-7820X with 3.6 GHz, the GPU is Nvidia GeForce GTX 1060 6 GB, and the RAM is 128 GB. All mosaic images are generated by Agisoft Metashape 1.65.

C. Experiment Results and Analysis

1) Accuracy of Estimated Irradiances: The DLS irradiance time series and estimated irradiance time series of three datasets are shown in Fig. 4. In general, the estimated irradiance time series exhibit undulations. For the irradiance data of a single strip, it is apparent that the image at the end of the strip has a large irradiance error, while the image in the middle of the strip, which is close to the reference image, has a smaller error. Thus, the undulation is caused by accumulated errors. Besides, the average relative error of the three datasets is 11.98%, 10.47%, and 12.88%, respectively.

2) Effectiveness of Illumination Compensation: For the HIT dataset, the mosaic images corrected by four methods are shown in Fig. 5(a)–(d) in true color with a 5% linear stretch. Although the 5% linear stretch exaggerates the error visually, some details of the results may not be apparent in true-color mosaic images. Then the raster color slices of NRMSE in reflectance factors between the corrected and reference mosaic image are shown in Fig. 5(e)–(h) to visualize the effectiveness of the four methods.

Similarly, the mosaic images corrected by four methods of the SKM dataset are shown in Fig. 6(a)–(d) in true color. The raster color slices of NRMSE in reflectance factors between the corrected image and the reference mosaic image are shown in Fig. 6(e)–(h). And the mosaic images corrected by four methods of the YRD dataset are shown in Fig. 7(a)–(d) in true color. The raster color slices of NRMSE in reflectance factors between the corrected image and the reference mosaic image are shown in Fig. 7(e)–(h).

The mean value and STD of NRMSE of the three datasets for the entire experiment scene are shown in Table I, in which the lowest result is displayed in bold.

We choose ELM [7] to represent errors before the illumination compensation as it is not robust to the illumination variation. In Figs. 5–7(a), there is a slight color difference in the area that appears deep red or slight green in Figs. 5–7(e). Fig. 5(e) shows that there are about three strips with a smaller NRMSE displayed in green and about five strips with a large NRMSE of around 15–25%. In Fig. 6(e), there are about two strips with a huge NRMSE of over 25%, four strips with a large NRMSE of around 8%–25%, and four strips with a small NRMSE. And Fig. 7(e) shows that there are about five strips with a large NRMSE of around 15–25% and three strips with a smaller NRMSE. The NRMSE is large when the irradiance of the corrected image and the reference mosaic differs a lot, and vice versa. Thus, the NRMSE of ELM corresponds with the variation of DLS irradiance in Fig. 3(d)–(f).

It can be seen in Figs. 5–7(f) that the NRMSE result is target-related. The NRMSE is large for targets with high and low reflectance, while it is small for targets with reflectance factors near the mean value of the subimage. It proves that GELM is more of an image homogenization method. GELM works badly on the HIT and SKM datasets due to the complex nature of the data scene and the abundant nature of the ground targets.

As shown in Figs. 5–7(c), RCOP works fairly well in the area near the calibrated tarp, but there are some color differences on the margin of the mosaic image caused by accumulated errors. Similarly, Figs. 5–7(g) shows that the NRMSE is only small for pixels that are not far away from the reference tarp, which corresponds to its principle of the optimal path. Besides, the NRMSE result of pixels in the margin area is above 20%. Although RCOP was proven to work well on several dozens of MSIs before [21], it does not work well on hundreds of MSIs due to the huge accumulated error.

As shown in Figs. 5–7(d) and Figs. 5–7(h), there are few color differences in the mosaic image, and the NRMSE result in most areas is below 8%, displayed in green. As shown in Table I, the proposed method can decrease the mean NRMSE introduced by illumination variations from around 20% to lower than 10%. However, there are still areas in the top left corner and bottom right corner with large errors in Fig. 5(h).

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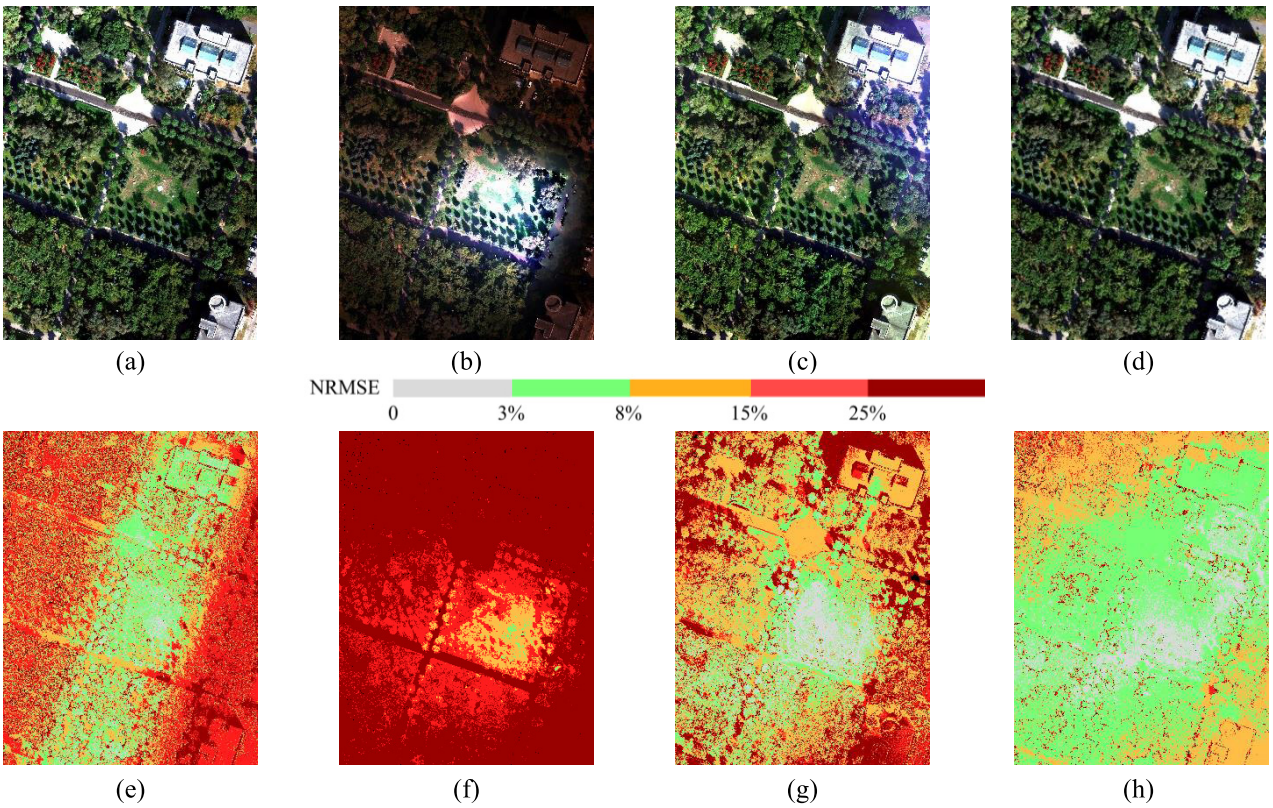


Fig. 5. (a)–(d) Mosaics of the HIT dataset of the ELM, GELM, RCOP, and our method in a 5% linear stretch. (e)–(h) Reflectance NRMSE between the reference mosaic image and corrected image by ELM, GELM, RCOP, and our method.

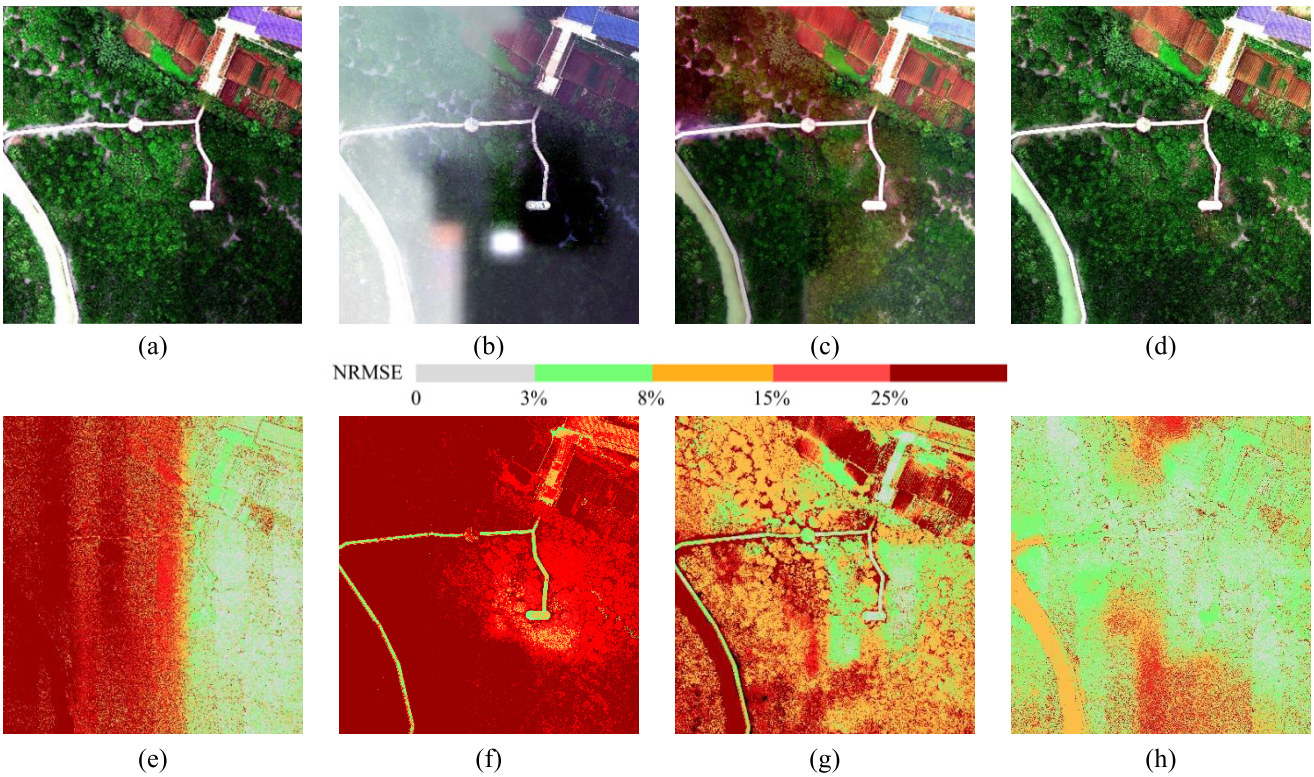


Fig. 6. (a)–(d) Mosaics of the SKM dataset of the ELM, GELM, RCOP, and proposed method in a linear stretch. (e)–(h) Reflectance NRMSE between the reference mosaic image and corrected image by ELM, GELM, RCOP, and our method.

It might be caused by the BRDF behavior of roads and roofs, which leads to the difference in reflectance factors of tie points, and there are still areas below the mosaic image with large errors in Fig. 6(h), which might be caused by the few tie

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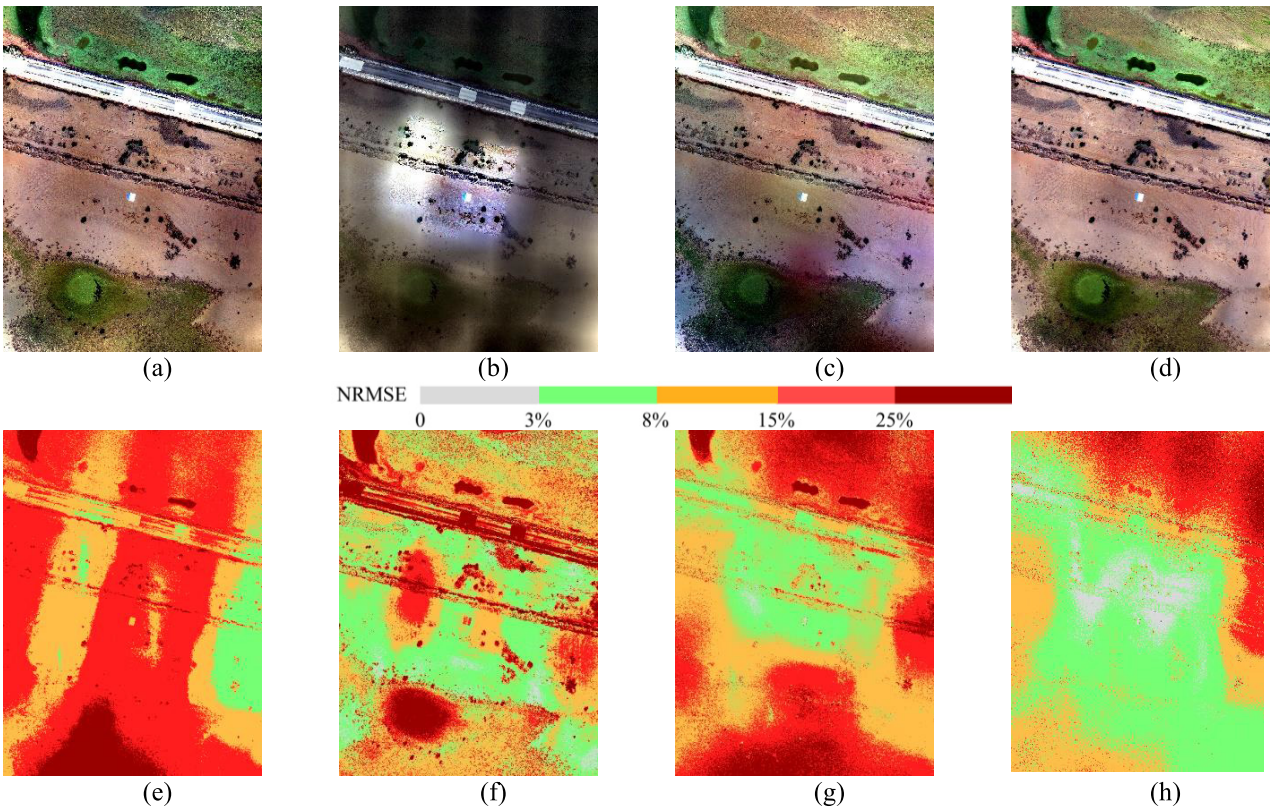


Fig. 7. (a)–(d) Mosaics of the YRD dataset of the ELM, GELM, RCOP, and our method in a 5% linear stretch. (e)–(h) Reflectance NRMSE between the reference mosaic image and corrected image by ELM, GELM, RCOP, and our method.

points. As for the result of the YRD dataset in Fig. 7(h), the NRMSE on the margin of the mosaic image is above 15%, which means that the influence of accumulated errors is huge in the YRD dataset.

This is mainly due to the obvious BRDF behavior of bare land, which is the primary land cover in the experimental scene. By the way, since the HIT dataset is a complex scene, there are large deviations between the edge pixels of targets of the corrected mosaic image and reference mosaic image. Thus, the result of the HIT dataset is visually better than that of the YRD dataset although their mean NRMSE results are similar.

It can be seen in Figs. 5–7(g) and (h) that the NRMSE results of the proposed method are much smaller than that of RCOP in the margin areas. It means that the accumulated errors of the proposed method are much smaller than that of RCOP because the irradiance of hundreds of MSIs is obtained in a single mathematical operation instead of stepwise iterations in the proposed method.

To sum up, the proposed method has a clear advantage for illumination compensation on hundreds of images without irradiance sensors. It works pretty well in forest scenes, such as the SKM dataset. It also works well in the urban forest, which contains many man-made BRDF targets by using qualified tie points. As for the scenes with obvious BRDF behaviors, the effectiveness of the proposed method is reduced but still higher than the existing methods. The good performance of the proposed method relies on its two advantages. The first one is that it compensates for the illumination variation with the physical imaging principle. We can achieve a good performance in alleviating the illumination variation with the estimated illumination irradiances. The second advantage is that it can minimize the accumulated errors on marginal images since the hundreds of irradiances are obtained in a single mathematical operation instead of stepwise iterations.

The proposed illumination estimation model is suitable for numerous MSIs and can estimate the irradiance of all images at the same time.

3) Accuracy of Reflectance Factors: As for the HIT dataset, we measured the reflectance factors of 13 black fabric, ten areas of roads, and eight carton boxes. Since there are some targets in shadow or not visible in MSIs due to the trees, we evaluate the accuracy of MSI with all the remaining ground truth (seven black fabric, eight roads, and four carton boxes). Similarly, for the SKM dataset, we measured the reflectance factors of 5 aerial survey targets as ground truth and evaluated the accuracy of MSIs with them. Because of the function of automatic adjustment of integration time, there is an overexposure phenomenon in 650–720 nm for aerial survey targets in high reflectance. So, we use the remaining six bands for evaluation.

The RMSE and NRMSE of ground truth targets on the HIT and SKM datasets are shown in Table II. For each type of ground truth, there are several targets close to the calibrated tarp (shading in white in Table II) and several targets far away from it (shading in light gray in Table II) to evaluate the effectiveness of illumination compensation in different areas.

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TABLE II
RMSE AND NRMSE OF GROUND-TRUTH TARGETS ON THE HIT CAMPUS AND SKM DATASET

Dataset	Targets	Num	DLS + Tarp		ELM		GELM		RCOP		Proposed Method	
			RMSE (%)	NRMSE (%)	RMSE (%)	NRMSE (%)	RMSE (%)	NRMSE (%)	RMSE (%)	NRMSE (%)	RMSE (%)	NRMSE (%)
HIT	Black Fabric (The reflectance is about 3~5%)	1	0.43	14.84	0.62	17.09	2.28	87.00	1.27	33.05	0.74	20.29
		2	0.65	14.96	0.90	20.21	1.95	43.86	1.23	28.04	0.91	21.81
		3	0.71	16.27	0.95	21.79	2.70	61.81	1.43	32.74	0.93	21.44
		4	0.41	13.90	1.27	43.01	2.28	65.08	1.21	35.92	0.98	26.21
		5	0.55	17,18	1.43	43.03	2.03	69.28	1.51	41.09	1.11	28.62
		6	0.93	21.47	2.04	46.80	3.55	81.47	2.21	50.72	1.19	27.59
		7	0.54	14.88	1.70	46.69	2.91	79.69	1.85	46.54	0.92	25.35
		Mean	**0.60**	**16.05**	**1.27**	**34.08**	**2.52**	**69.74**	**1.57**	**37.72**	**0.96**	**24.47**
	Roads (The reflectance is about 20%)	1	1.10	7.61	1.62	9.89	11.65	75.66	1.87	12.15	1.64	10.67
		2	1.39	9.10	1.70	11.04	11.33	73.57	1.77	11.52	1.79	11.66
		3	1.35	7.03	3.51	15.87	10.76	56.88	3.20	16.95	2.06	10.89
		4	1.28	6.79	4.02	21.285	11.52	60.96	3.34	17.66	2.01	10.67
		5	1.26	6.19	4.695	22.98	12.52	65.89	3.36	16.44	2.34	11.46
		6	1.35	6.59	4.38	21.48	13.41	70.89	3.47	16.97	1.94	9.49
		7	2.47	9.88	5.30	21.55	20.21	80.76	7.19	28.75	2.68	10.16
		8	1.86	10.5	5.25	29.68	15.87	89.93	5.29	29.97	2.72	15.46
		Mean	**1.50**	**8.01**	**3.90**	**19.91**	**13.40**	**71.81**	**3.93**	**20.39**	**2.38**	**12.81**
	Carton Boxes (The reflectance is about 35%)	1	2.33	8.05	3.42	10.43	8.60	31.59	3.20	11.06	2.13	7.38
		2	3.39	9.91	4.64	13.48	18.29	56.70	6.06	17.72	4.67	13.30
		3	2.98	9.36	6.36	19.94	17.79	51.59	7.58	22.00	5.51	15.63
		4	3.94	11.98	5.37	23.39	18.36	55.88	8.48	27.95	4.67	19.59
		Mean	**3.16**	**9.82**	**4.94**	**16.81**	**15.76**	**48.94**	**6.33**	**19.68**	**4.24**	**13.97**
SKM	Aerial Survey Targets (The reflectance is about 80%)	1	7.21	11.79	8.75	14.68	19.67	35.44	11.27	18.42	8.90	15.44
		2	8.45	13.82	10.32	16.53	20.53	33.07	9.87	16.14	10.01	17.37
		3	10.20	16.68	20.36	33.31	29.76	48.67	18.99	31.05	10.96	19.02
		4	10.28	16.80	18.03	29.49	30.11	49.24	17.51	28.63	9.00	15.63
		5	9.36	15.30	15.34	25.19	29.53	48.30	15.80	25.84	12.29	21.33
		Mean	**9.1**	**14.878**	**14.56**	**23.84**	**25.92**	**42.94**	**14.68**	**24.016**	**10.23**	**17.75**

As for the ELM, the error of the targets close to the tarp is small, and the error of the target far away from the tarp is large, which is caused by the varying illumination. As shown in Table II, there is no difference between the errors of targets far away from the tarp of RCOP and those of ELM, which means that the negative effects of accumulated errors can almost obscure the effort of RCOP. Besides, the proposed method outperforms current methods, and the NRMSE on three datasets is noticeably reduced. In the absence of irradiance sensors, the effectiveness of illumination compensation of our method is close to that of the irradiance-sensor-oriented method.

D. Parameters Analysis

The first parameter of the proposed method is η mentioned in Section II-A. It is used to filter the qualified tie points, which do not behave with obvious BRDF effect, so as to reduce the impact of accumulated errors in the corrected Mosaic images. The value of η is set to make sure the number of pairs of tie points between overlapping images is around 50. The η of the three datasets are shown in Table III.

For analyzing the influence of the parameter α, the mean value of NRMSE with respect to changes of α on the three datasets is shown in Fig. 8.

TABLE III
VALUES OF PARAMETER η OF THREE DATASETS

Dataset	HIT	SKM	YRD
η	0.8	1.0	0.75

TABLE IV
TIME COMPLEXITY AND MEMORY CONSUMPTION OF THREE METHODS

		RCOP	GELM	Proposed Method
Time Complexity		$O(N^2P)$	$O(mN^2P)$	$O(mN^2P^3)$
Memory Consumption	HIT		1.06G	5.88G
	SKM		0.52 GB	2.52 GB
	YRD		6.2 GB	15.8 GB

As shown in Fig. 8, the mean value of NRMSE on each dataset decreases slightly at first and finally increases with the increase of α. Besides, the mean NRMSE result of the three datasets becomes relatively optimal at the turning

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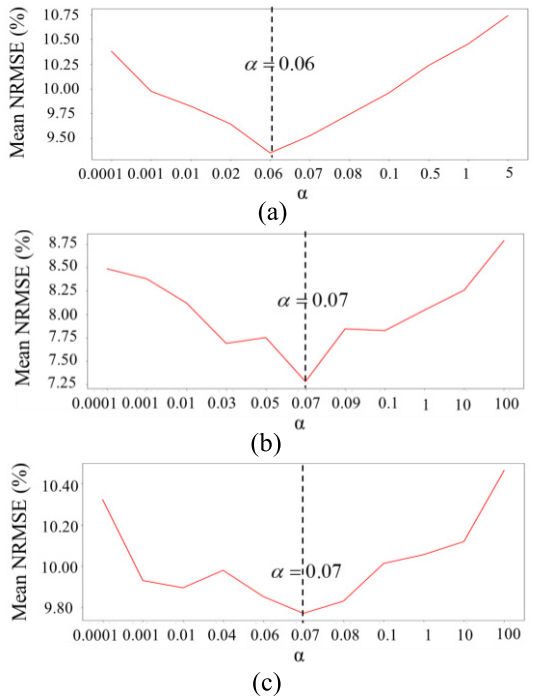


Fig. 8. Mean NRMSE with respect to changes of α on the three datasets. (a) HIT. (b) SKM. (c) YRD.

point where α is equal to 0.06, 0.07, and 0.07, respectively. Therefore, the optimal values of α on three different datasets are relatively stable, which represents the ratio of importance weights between the priors on reference points and tie points. Considering that the mean NRMSE does not respond greatly to changes in α, it is reasonable to generalize the empirical value of α to the other datasets, which is 0.07.

E. Time Complexity and Memory Consumption

The tie-point extraction is a common source of the time complexity for RCOP, GELM, and the proposed method. It takes about 1–5 s for each pair of overlapping images. Thus, we only analyze the time complexity of solving the linear coefficients or illumination irradiances. Let N denote the number of images, P denote the band number of MSI, and m denote the number of pairs of tie points. The time complexities of the three methods are shown in Table IV.

The main source of memory consumption is the storage of tie points. There is no extra consumption of memory for RCOP since images are corrected image by image. Both the GELM and proposed method need tie points of all images to estimate the illumination irradiance of all MSIs simultaneously, which requires extra memory consumption. The memory consumption of GELM and the proposed frame on three datasets are also shown in Table III.

It can be seen that the memory consumption of the proposed framework is several times of GELM because the former process the images of all bands simultaneously. The experimental results in Section III-C show that the proposed method can achieve good performance on the illumination compensation but suffer from more memory complexity.

Besides, since the overlapping rates of the HIT and SKM datasets are both 60% while that of the YRD dataset is 85%, the number of overlapping images pair of YRD is several times of the other two datasets. Thus, the memory consumption of the YRD dataset is much higher.

IV. Conclusion

This article proposed a physics-based illumination estimation and compensation method for numerous MSIs without irradiance sensors. The key idea is to substitute the irradiance sensor measurement with the estimated irradiance. In the proposed method, we first formulated the illumination estimation model based on the intraimage hypothesis on illumination consistency and the interimage hypothesis on reflectance consistency. The irradiances of numerous MSIs were estimated simultaneously with the estimation model. To our knowledge, this is the first attempt to estimate the illumination irradiance for MSIs simultaneously. Then based on the physical imaging principle, the influence of the varying illumination can be alleviated with the estimated irradiance. The proposed method is validated on three field-experiment datasets. Besides, the time and memory complexity of the proposed method are discussed. The results indicate good performance on the illumination compensation but suffer from more memory complexity.

For future work, it is meaningful to further consider the influence of the BRDF effect in the proposed method, as it still has large accumulated errors in scenarios with obvious BRDF behavior.

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